Performance characteristics of the DPMs

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1 Methods

1.1 Parametric DPM

We denote the biomarker measurement k for individual i at time j as y_{ijk}, where $i = 1, \ldots, n$, $k = 1, \ldots, p$ and $j = 1, \ldots, q_{ik}$. The model can be written as:

$$y_{ijk} = f(s_{i,j}, \Theta_k) + e_{ijk},$$

where f corresponds to the model, $z_{i,j}$ are latent random variables associated to the i-th subject, Θ_k are the model parameters, which include variables from f and $s_{i,j}$, and e_{ijk} is an error term.

In general, the aim is to identify the parameters that most accurately represent the observations, such as those that optimize the likelihood [**?**] or optimization the robust nonlinear regression [**?**].

1.2 RPDPM

The latent variable $s_{i,j}$ transforms the patient's age $t_{i,j}$ into a timescale that defines the progression of the disease, where α_i and β_i represent the temporal parameters for the i-th patient:

$$s_{i,j} = \alpha_i \cdot t_{i,j} + \beta_i.$$

The geometric model for the trajectories is defined by:

$$f(s; \Theta) = (a - d)g(s; \Theta) + d$$

where a and d are normalization parameters for the markers, and $g(s; \Theta)$ represents a logistic function. Finally, the multiobjective optimization for robust nonlinear regression is defined as:

$$\left\{\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\Theta}}\right\} = \min_{i,j,k} \sum_{i,j,k} \omega_i \rho \left(\frac{y_{ijk} - f(s_{i,j}, \Theta_k)}{\sigma_k} \right)$$

where $\rho(\cdot)$ is a maximum likelihood-type function and $\omega_i = 1/N_i$ is a weighting factor for normalizing the objective function with the number of available points per subject (N_i).

1.3 Leaspy

2 DPM's performances

Assessing the performance of DPM is essential. This evaluation can be carried out through two types of measures: a) Measuring the accuracy in estimating the temporal trajectories of patients' markers. b) Evaluating patients' diagnosis over time, including estimates of remaining time for key clinical events, such as the age at which cognitive decline begins.

2.1 Evaluations of estimated trajectories

The Mean Absolute Error (MAE) is a performance measure for models that is less sensitive to outliers (Chai and Draxler, 2014), calculated based on the absolute differences between actual and estimated values:

$$MAE_k = \frac{1}{N_k} \sum \left| y_{y,j,k} - \hat{y}_(i,j,k) \right|$$

N_k represents the total number of measurements for the k-th marker across all subjects and visits. The measure and estimated value of this marker for the i-th subject at the j-th visit are denoted by $y_{i,j,k}$ and $\hat{y}(i,j,k)$ respectively. To enable cross-variable comparisons, the MAEs are normalized as NMAE [93], as in Equation (13)

2.2 Clinical status classification

$s_{i,j}$ is the disease progression score (DPS) for the j-th visit of the i-th subject and is obtained as:

$$s_{i,j} = \alpha_i \cdot t_{i,j} + \beta_i,$$

where $t_{i,j}$ is the age of subject i in visit j, and α_i and β_i are the rate and onset of disease progression of subject i, respectively.

The clinical status is classified based on the DPSs with a Bayesian classifier that uses the kernel density estimation (KDE) (Parzen, 1962)-based fitted likelihoods as:

$$p(s|c) = \frac{1}{n \cdot h} \sum_{t=1}^{n} K\left(\frac{s - s_t}{h}\right)$$

where $K(\cdot)$ is the kernel, $h \geq 0$ is a smoothing parameter and c denotes a clinical group. The predicted class for observation $s_{i,j}$ is c_r, which maximizes the posterior probability:

$$r = \arg\max_r \left(p(c_r) \cdot p(s_{i,j}|c_r) \right).$$

2.3 Measures regarding the timeframes of conversion toward cognitive decline

Quality measures were proposed for DPMs associated with a specific clinical event: in this case, the onset of cognitive decline in a patient. Firstly, the estimated age at which the subject initiates this decline was calculated according to a proposed model. This date was determined as the average time between the first visit diagnosed with MCI and the preceding visit where the patient was considered to have normal cognitive function. Additionally, the patient's final visit in the

longitudinal study must confirm the patient's cognitive decline or dementia status. Based on the proposed date for this clinical event in the i-patient, denoted as $Age_{i_{onset}}$, the following measures are suggested.

For sCU subjects, the age at the event must be greater than the patient's age at the last visit:

$$SCORE_1 = \frac{\#\{i|(i \in sCU) \cap ((Age_{i_{end}} < Age_{i_{onset}})\}}{\#sCU}$$

For sMCI subjects, the age at event must be the same as the age at baseline:

$$SCORE_2 = \frac{\#\{i|(i \in sMCI) \cap ((Age_{i_1} < Age_{i_{onset}})\}}{\#sMCI}$$

Regarding subjects progressing from CU to MCI, we measured: a) the percentage of preclinical CU cases detected by the DPMs:

$$SCORE_3 = \frac{\#\{i|(i \in sMCI) \cap ((Age_{i_1} < Age_{i_{onset}})\}}{\#pCU}$$

b) the correlation between age at the event and its estimation by the model, and c) the correlation between the years remaining for the individual to reach cognitive decline according to clinical diagnosis and the model's estimation from the visit's date at baseline.

2.4 Data partitions and boostraping

To assess the algorithms, the dataset is divided into two non-overlapping sets for training and testing. Furthermore, bootstrapping is employed in the experiments to evaluate the DPM's performances. Within each bootstrap iteration, a subset of the training subjects is randomly sampled with replacement, taking into account their clinical grouping, which ensures that each bootstrap sample includes data representing all diagnostic statuses.